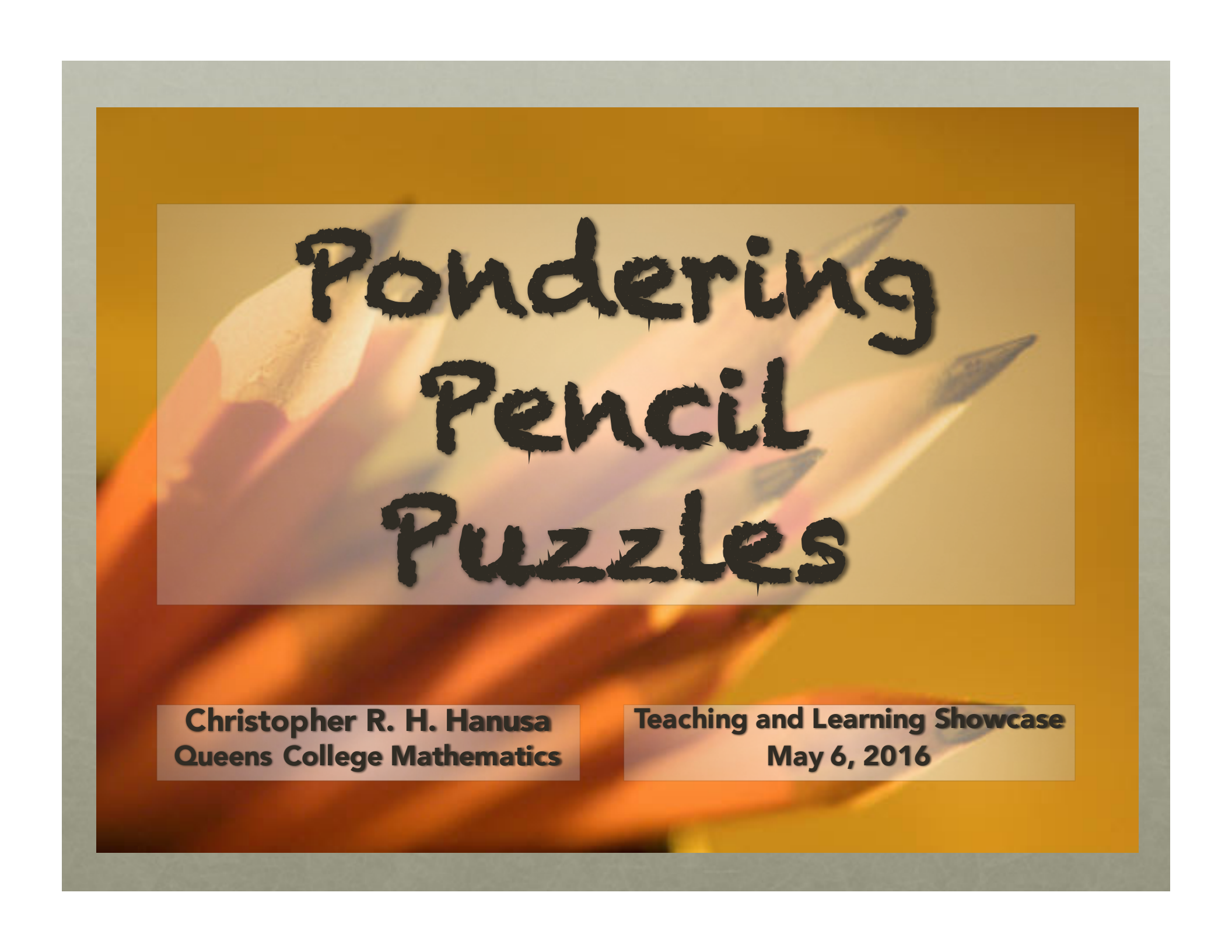
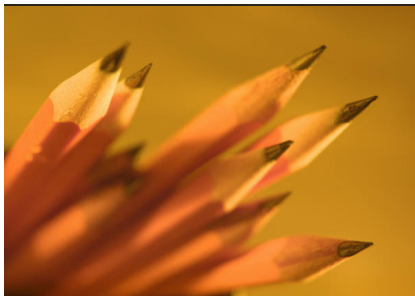


The background of the slide features a warm, orange-toned image of several hands holding pencils in various orientations, creating a sense of collaborative learning and problem-solving.

Pondering Pencil Puzzles

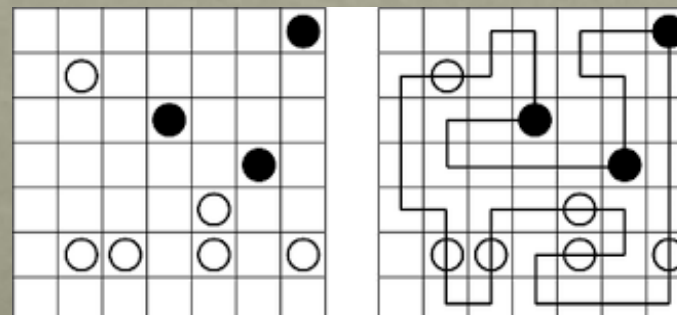
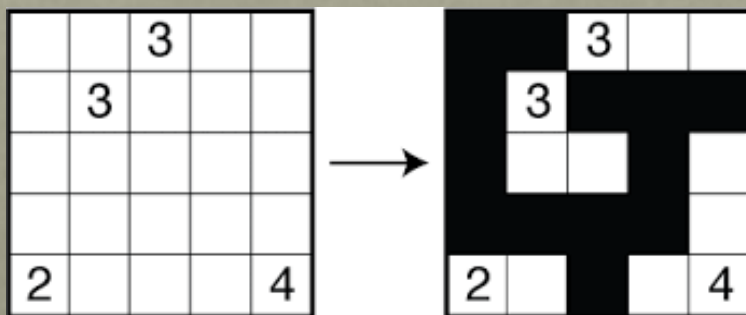
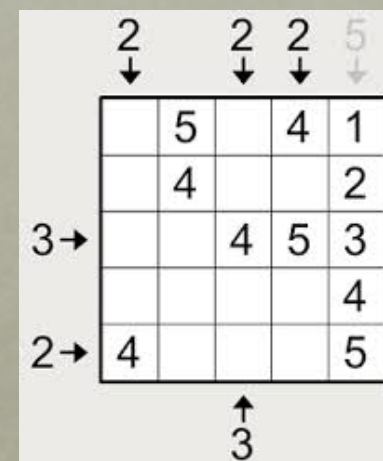
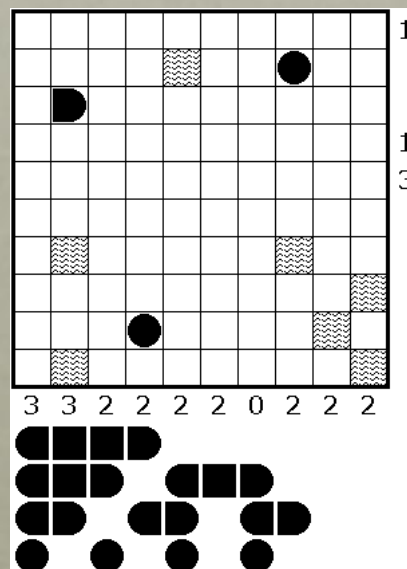
Christopher R. H. Hanusa
Queens College Mathematics

Teaching and Learning Showcase
May 6, 2016



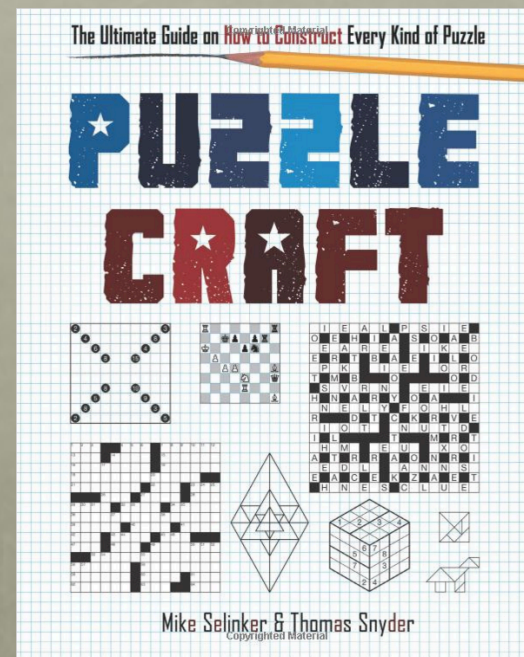
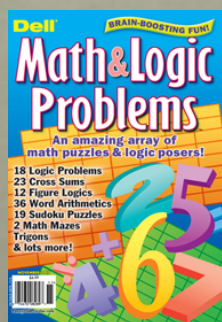
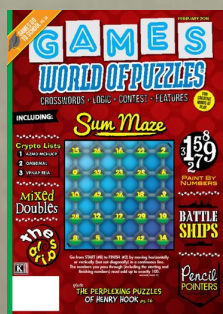
Pencil Puzzles?

- Most well known: **Sudoku**
- Diversity of types!
- Pure logic – no "math"
 - Nikoli
 - Conceptis Puzzles
 - Puzzle Picnic



The Course

- **MATH 555: Games and Puzzles**
 - Mon/Wed 75 min
 - 11 undergraduates
- **Textbook?**
 - Magazines
 - Puzzlecraft





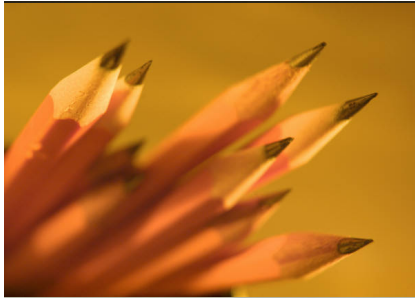
Course Goals

- Learn problem solving techniques
- Design own puzzles
- Develop mathematical curiosity
- Become a puzzle connoisseur



HARD!

A chance to **EXPERIENCE** and
BE CREATIVE WITH mathematics



Certified Organic

- **Course content determined organically with input from students**
- **How would I assign a grade?**
 - Exams?!?
 - Solving prowess?
 - Puzzle creation? ✓
 - Essays? ✓
 - Wikis?
 - Final project? ✓



Daily Format

How to teach puzzles?

1. Introduce the rules
2. Solve some puzzles of varying difficulty
3. Discuss solving strategies
 - In-class scribe
 - Real-time puzzle solving
4. Try to create puzzles
 - At home / In groups



Daily Format

- 5. Solve, critique others' puzzles**
- 6. Philosophical discussions**
 - What makes a puzzle hard?
 - Tension between difficulty and uniqueness.
 - How much information is in a clue?
 - Theory: decision tree
- 7. Homework Assignment**
- 8. Presentation**



Assignments

Creative Puzzling Assignment

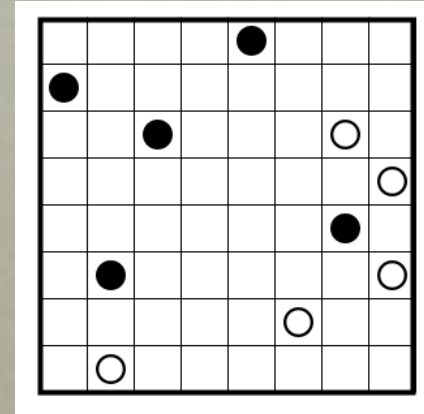
- **Create one or more puzzles**
 - Push the boundaries
- **Write a short essay (Deep thought!)**
 - What conscious decisions went into its construction?
 - What makes the puzzle easier (harder) to create? solve?
 - Explain the tension between uniqueness and difficulty.

Inspired by Circular Teaching Squad discussion

- **Join us! Watch for announcements on Socrates**

Fun Lessons

- What are the rules of this game?
- Create a hybrid puzzle.
 - For two puzzles you've learned
 - Merge rules and clues
- Voice record yourself completing the puzzle.





Field Trip!

- **An Escape Room**
 - Sunday in Flushing
- **Locked in a room!**
 - Escape within an hour!
- **Solve a sequence of puzzles**
 - Find hidden objects
 - Code breaking
 - Temporal, physical clues
 - (hints provided)

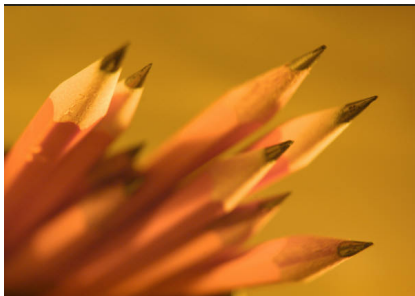




Final Project

- **Create an “Escape the Room”**
 - Group theme
 - Individual puzzles + writeup
 - Group portfolio: Walkthrough
- **Final exam day:**
 - SOLVE the other group’s room
- **Student-designed assignment**
 - Students taking ownership





Student Comments

Looking into the why of a puzzle was something I never thought about prior to the course. I feel like a better puzzle solver now!

Creativity played a big role in this course which forced me to get out of my usual structured mindset.

A very open class that will bend your mind a bit.

It's not often in math you're asked to really dive into why you're doing something.

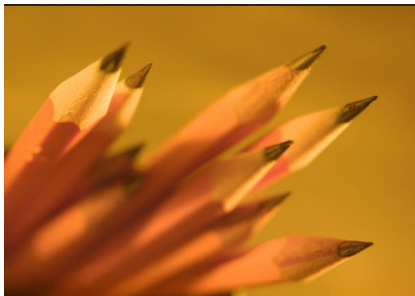
Math 555 has definitely shaped how I approach my other classes. I found myself looking at other mathematical problems from an alternative perspective.



TIL

(things I learned)

- **You can create a puzzle in a group!**
- **Very useful: Pictures before & after getting stuck**
- **Change in solving philosophy**
 - **Solve the puzzle with an eye toward creation**
- **I gave up control and I feel fine!**
 - **Came out of class with a different homework assignment than when I went in.**
- **Supervising 35 projects in a semester is tiring.**



THANK YOU!

qc.edu/~chanusa

> **Course Archive**
Course Content

> **Talks**
Slides Available

- **QC Mathematics Dept**
- **Sarah Mason (Wake Forest)**
- **My students, who amaze and inspire, EVERY TIME!**



Christopher R. H. Hanusa Queens College, CUNY

Teaching and Learning Showcase May 6, 2016